EE 66 Signals, Dynamics, and Information
Fall 2026 Homework 1

This homework is due Friday, September 4th, 2026, at 23:59.

Submission Format
Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned).

Submit the file to the appropriate assignment on Gradescope.

1. Complex Proofs (Adapted from EE 120 Homework)

Consider a pair of complex numbers v and z. Prove each of the following assertions.

(a) $z+\bar{z}=2\Re(z)$, where $\Re(z)$ denotes the real part of z.

Solution:
$z+\bar{z}=(a+bi)+(a-bi)=2a$

(b) Let $z=a+ib$, where $a,b\in\mathbb{R}$. Then $z\bar{z}\geq 0$, with equality if, and only if, $z=0$.

Solution:
$z\bar{z}=(a+ib)(a-ib)=a^2+b^2\rightarrow a^2+b^2\geq 0$
Because $a,b\in\mathbb{R}$, $a^2,b^2\geq 0$.
The sum of two positive numbers must be greater than or equal to the numbers in question, so if either a or b is non-zero (their square is greater than 0), then the entire sum must be greater than zero. Therefore, both a and b must be zero, $z=a+ib=0+0i=0$ in order for $z\bar{z}=0$.

(c) z is real if, and only if $z=\bar{z}$.

Solution:
$z\in\mathbb{R}\iff z=a+0i=a-0i=\bar{z}$

(d) $\overline{(zv)}=\bar{z}\bar{v}$.

Solution:
Say $z=a+bi$ and $v=c+di$.
$\overline{(zv)}=\overline{[(a+bi)(c+di)]}=\overline{(ac+adi+bci+bdi^2)}=\overline{(ac+adi+bci+bd(-1))}$
$=\overline{[(ac-bd)+(ad+bc)i]}=(ac-bd)-(ad+bc)i$
$\bar{z}\bar{v}=\overline{(a+bi)}\,\overline{(c+di)}=(a-bi)(c-di)=ac-adi-bci+bdi^2=(ac-bd)-(ad+bc)i$
Therefore $\overline{(zv)}=\bar{z}\bar{v}$

2. Vector Basics

Evaluate the following expressions:

(a) In $\mathbb{R}^3$, evaluate $\mathbf{e_1} - 3\mathbf{e_2} - 4\mathbf{e_3}$

Solution: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - 3\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - 4\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ -4 \end{bmatrix}$

(b) $\mathbf{1}^\top \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}$

Solution: $a+b+c+d+e$

(c) $\mathbf{x}^\top\mathbf{x}$, where $\mathbf{x} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

Solution: $a^2+b^2+c^2$

(d) $(\mathbf{e_1}+\mathbf{e_2})^\top\mathbf{x}$, where $\mathbf{x} = \begin{bmatrix} 0.5i \\ 3.7 \\ 4.3+3i \end{bmatrix}$

Solution: $0.5i+3.7$

(e) $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}^\top \begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}$

Solution: $a+d$

3. Inner Product Properties

Our definition of the inner product in $\mathbb{R}^n$ is:

$$\langle \mathbf{x}, \mathbf{y} \rangle \;=\; x_1 y_1 + x_2 y_2 + \ldots + x_n y_n \;= \mathbf{x}^\intercal \mathbf{y}\ , \quad \text{for any } \ \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Prove the following identities in $\mathbb{R}^n$:

(a) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$

Solution: This is seen by direct expansion:
Let $x_i, y_i \in \mathbb{R}$, then

$$\begin{aligned}
\left\langle \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \right\rangle &= x_1 \cdot y_1 + x_2 \cdot y_2 + \cdots + x_n \cdot y_n \\
&= y_1 \cdot x_1 + y_2 \cdot x_2 + \cdots + y_n \cdot x_n \\
&= \left\langle \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right\rangle
\end{aligned}$$

So the inner product is commutative.

(b) $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$
Note: $||\cdot||$ in this case indicates the Euclidean norm, which is defined as $||\mathbf{x}|| = \sqrt{x_1^2 + x_2^2 \ldots + x_n^2}$

Solution:

$$\begin{aligned}
\left\langle \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right\rangle &= x_1 \cdot x_1 + x_2 \cdot x_2 + \cdots + x_n \cdot x_n \\
&= x_1^2 + x_2^2 + \cdots + x_n^2 \\
&= (\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2})^2 \\
&= (\|\mathbf{x}\|)^2
\end{aligned}$$

The inner product of a vector with itself is its norm squared.

(c) $\langle -\mathbf{x}, \mathbf{y} \rangle = -\langle \mathbf{x}, \mathbf{y} \rangle$.

Solution:

$$\begin{aligned}\langle -\mathbf{x}, \mathbf{y}\rangle &= \left\langle \begin{bmatrix} -x_1 \\ -x_2 \\ \vdots \\ -x_n \end{bmatrix}, \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \right\rangle \\ &= -x_1 \cdot y_1 - x_2 \cdot y_2 - \cdots - x_n \cdot y_n \\ &= -(x_1 \cdot y_1 + x_2 \cdot y_2 + \cdots + x_n \cdot y_n) \\ &= -\langle \mathbf{x}, \mathbf{y}\rangle\end{aligned}$$

Flipping the sign of one of the vectors in the inner product flips the sign of the inner product, but does not change the magnitude.

(d) $\langle \mathbf{x}, \mathbf{y} + \mathbf{z}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{x}, \mathbf{z}\rangle$

Solution:

$$\begin{aligned}\langle \mathbf{x}, \mathbf{y} + \mathbf{z}\rangle &= \mathbf{x}^\mathsf{T}(\mathbf{y} + \mathbf{z}) \\ &= \mathbf{x}^\mathsf{T}\mathbf{y} + \mathbf{x}^\mathsf{T}\mathbf{z} \\ &= \langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{x}, \mathbf{z}\rangle\end{aligned}$$

The inner product is distributive.

(e) $\langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{x}\rangle + 2\langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{y}, \mathbf{y}\rangle$

Solution: Using the distributive and commutative properties from the previous parts,

$$\begin{aligned}\langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}\rangle &= \langle \mathbf{x}, \mathbf{x} + \mathbf{y}\rangle + \langle \mathbf{y}, \mathbf{x} + \mathbf{y}\rangle \\ &= \langle \mathbf{x}, \mathbf{x}\rangle + \langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{y}, \mathbf{x}\rangle + \langle \mathbf{y}, \mathbf{y}\rangle \\ &= \langle \mathbf{x}, \mathbf{x}\rangle + 2\langle \mathbf{x}, \mathbf{y}\rangle + \langle \mathbf{y}, \mathbf{y}\rangle\end{aligned}$$

4. Subspaces Practice

For each of the sets $\mathbb{U}$ (which are *subsets* of $\mathbb{R}^3$) defined below, state whether $\mathbb{U}$ is a *subspace* of $\mathbb{R}^3$ or not.

Note:

- To show $\mathbb{U}$ is a subspace, show that it satisfies all of the necessary properties of a subspace.
- To show $\mathbb{U}$ is not a subspace, you merely have to show that at least one of the properties fails- perhaps through a counterexample.

(a) $\mathbb{U} = \left\{ \begin{bmatrix} 2(x+y) \\ x \\ y \end{bmatrix} \mid x, y \in \mathbb{R} \right\}$

Solution: We test the three properties of a subspace:

i. Let $\mathbf{v_1} = \begin{bmatrix} 2(x_1+y_1) \\ x_1 \\ y_1 \end{bmatrix}$ be a member of the set $\mathbb{U}$. Assume $\mathbf{v_2} = \alpha \mathbf{v_1}$, where α is a scalar. Here

$$\mathbf{v_2} = \alpha \mathbf{v_1} = \alpha \begin{bmatrix} 2(x_1+y_1) \\ x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 2(\alpha x_1 + \alpha y_1) \\ \alpha x_1 \\ \alpha y_1 \end{bmatrix} = \begin{bmatrix} 2(x_i+y_i) \\ x_i \\ y_i \end{bmatrix},$$

where $x_i = \alpha x_1$ and $y_i = \alpha y_1$. Hence, $\mathbf{v_2} = \alpha \mathbf{v_1}$ is a member of the set as well and the set is closed under scalar multiplication.

ii. Let $\mathbf{v_1} = \begin{bmatrix} 2(x_1+y_1) \\ x_1 \\ y_1 \end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix} 2(x_2+y_2) \\ x_2 \\ y_2 \end{bmatrix}$ be members of the set $\mathbb{U}$. Now, let us assume $\mathbf{v_3} = \mathbf{v_1} + \mathbf{v_2}$:

$$\mathbf{v_3} = \mathbf{v_1} + \mathbf{v_2} = \begin{bmatrix} 2(x_1+y_1) + 2(x_2+y_2) \\ x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} = \begin{bmatrix} 2(x_1+x_2+y_1+y_2) \\ x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} = \begin{bmatrix} 2(x_3+y_3) \\ x_3 \\ y_3 \end{bmatrix},$$

where $x_3 = x_1 + x_2$ and $y_3 = y_1 + y_2$ Hence, $\mathbf{v_3}$ is a member of the set as well and the set is closed under vector addition.

iii. Let $\mathbf{v_0} = \begin{bmatrix} 2(x_0+y_0) \\ x_0 \\ y_0 \end{bmatrix}$ be a member of the set, where we choose $x_0 = 0$ and $y_0 = 0$. So the vector $\mathbf{v_0} = \begin{bmatrix} 2(0+0) \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$. So the zero vector is contained in this set.

Hence we can decide that $\underline{\mathbb{U} \text{ is a subspace of } \mathbb{R}^3}$.

(b) $\mathbb{U} = \left\{ \begin{bmatrix} x \\ y \\ z+1 \end{bmatrix} \mid x, y, z \in \mathbb{R} \right\}$

Solution:

Again we check the three properties of a subspace:

i. Now let $\mathbf{v_1} = \begin{bmatrix} x_1 \\ y_1 \\ z_1+1 \end{bmatrix}$ be a member of the set $\mathbb{U}$. Assume $\mathbf{v_2} = \alpha\mathbf{v_1}$, where α is a scalar. Here

$$\mathbf{v_2} = \alpha\mathbf{v_1} = \begin{bmatrix} \alpha x_1 \\ \alpha y_1 \\ \alpha z_1+\alpha \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha y_1 \\ (\alpha z_1+\alpha-1)+1 \end{bmatrix} = \begin{bmatrix} x_i \\ y_i \\ z_i+1 \end{bmatrix},$$

where $x_i = \alpha x_1$, $y_i = \alpha y_1$ and $z_i = \alpha z_1 + \alpha - 1$. Hence, $\mathbf{v_2} = \alpha\mathbf{v_1}$ is a member of the set as well and the set is closed under scalar multiplication.

ii. Let $\mathbf{v_1} = \begin{bmatrix} x_1 \\ y_1 \\ z_1+1 \end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix} x_2 \\ y_2 \\ z_2+1 \end{bmatrix}$ be members of the set $\mathbb{U}$. Now, let us assume $\mathbf{v_3} = \mathbf{v_1} + \mathbf{v_2}$:

$$\mathbf{v_3} = \mathbf{v_1} + \mathbf{v_2} = \begin{bmatrix} x_1+x_2 \\ y_1+y_2 \\ z_1+z_2+2 \end{bmatrix} = \begin{bmatrix} x_1+x_2 \\ y_1+y_2 \\ (z_1+z_2+1)+1 \end{bmatrix} = \begin{bmatrix} x_3 \\ y_3 \\ z_3+1 \end{bmatrix},$$

where $x_3 = x_1+x_2$, $y_3 = y_1+y_2$ and $z_3 = z_1+z_2+1$. Hence, $\mathbf{v_3}$ is a member of the set as well and the set is closed under vector addition.

iii. Let $\mathbf{v_0} = \begin{bmatrix} x_0 \\ y_0 \\ z_0+1 \end{bmatrix}$ be a member of the set, where we choose $x_0 = 0$, $y_0 = 0$ and $z_0 = -1$. So the vector $\mathbf{v_0} = \begin{bmatrix} 0 \\ 0 \\ -1+1 \end{bmatrix} = \mathbf{0}$. So the zero vector is contained in this set.

Hence we can decide that $\mathbb{U}$ is a subspace of $\mathbb{R}^3$.

(c) $\mathbb{U} = \left\{ \begin{bmatrix} x \\ y \\ x+1 \end{bmatrix} \mid x,y \in \mathbb{R} \right\}$

Solution: Again we check the three properties of a subspace:

i. Now let $\mathbf{v_1} = \begin{bmatrix} x_1 \\ y_1 \\ x_1+1 \end{bmatrix}$ be a member of the set $\mathbb{U}$. Assume $\mathbf{v_2} = \alpha\mathbf{v_1}$, where α is a scalar. Here

$$\mathbf{v_2} = \alpha\mathbf{v_1} = \begin{bmatrix} \alpha x_1 \\ \alpha y_1 \\ \alpha x_1+\alpha \end{bmatrix} \neq \begin{bmatrix} x_i \\ y_i \\ x_i+1 \end{bmatrix},$$

where $x_i = \alpha x_1$ and $y_i = \alpha y_1$. Hence, $\mathbf{v_2} = \alpha\mathbf{v_1}$ is not a member of the set and the set is not closed under scalar multiplication.

ii. Let $\mathbf{v_1} = \begin{bmatrix} x_1 \\ y_1 \\ x_1+1 \end{bmatrix}$ and $\mathbf{v_2} = \begin{bmatrix} x_2 \\ y_2 \\ x_2+1 \end{bmatrix}$ be members of the set $\mathbb{U}$. Now, let us assume $\mathbf{v_3} = \mathbf{v_1} + \mathbf{v_2}$:

$$\mathbf{v_3} = \mathbf{v_1} + \mathbf{v_2} = \begin{bmatrix} x_1+x_2 \\ y_1+y_2 \\ x_1+x_2+2 \end{bmatrix} \neq \begin{bmatrix} x_3 \\ y_3 \\ x_3+1 \end{bmatrix},$$

where $x_3 = x_1+x_2$, and $y_3 = y_1+y_2$. Hence, $\mathbf{v_3}$ is not a member of the set and the set is not closed under vector addition.

iii. Let $\mathbf{v_0} = \begin{bmatrix} x_0 \\ y_0 \\ x_0 + 1 \end{bmatrix}$ be a member of the set. The first and third elements cannot both be zero regardless of the value chosen for x_0. So the zero vector is not contained in this set.

Hence we can decide that $\underline{\mathbb{U} \text{ is not a subspace of } \mathbb{R}^3}$. **Note that for full credit you only have to show that one of the properties is violated, you don't have to show all three.**

5. Inner Products and Orthogonality

For the purposes of this question, an inner product is defined as follows:

$$\langle \mathbf{x}, \mathbf{y} \rangle \;=\; x_1 y_1 + x_2 y_2 + \ldots + x_n y_n \;= \mathbf{x}^\mathsf{T} \mathbf{y} \;, \quad \text{for any } \; \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$

Show in $\mathbb{R}^2$ if two vectors are perpendicular, then they have an inner product of 0. You may **not** use the following definition: $\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta$.

Hint: The slope of a perpendicular line will be the negative reciprocal of the original line's slope. Therefore if the first vector is in the form of $\begin{bmatrix} a \\ b \end{bmatrix}$*, what must the perpendicular vector look like?*

Solution: Given the hint, we know that our two vectors will be of the form $\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\mathbf{y} = c \begin{bmatrix} -b \\ a \end{bmatrix}$, for some real constant c. Using the property of inner products that $\langle \mathbf{x}, c\mathbf{y} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle$, we can see the following:

$$\langle \mathbf{x}, c\mathbf{y} \rangle = c\langle \mathbf{x}, \mathbf{y} \rangle = c(-ba + ab) = 0$$

6. Polynomial Spaces

(a) Show that the set of all polynomials of the form

$$p(t) = p_0 + p_1 t + p_2 t^2 + p_3 t^3,$$

where $t \in \mathbb{R}$ and the coefficients p_k are real scalars, forms a vector space. Call this vector space $\mathbb{V}$. *Note:* Any of the coefficients p_k can be zero.

Solution:

No escape (scaling) property:

$$\alpha p(t) = \alpha p_0 + \alpha p_1 t + \alpha p_2 t^2 + \alpha p_3 t^3$$

The function above is, itself, a cubic polynomial, so the no escape property holds.

No escape (addition) property:
Let us define two cubic polynomials:

$$p(t) = p_0 + p_1 t + p_2 t^2 + p_3 t^3$$
$$q(t) = q_0 + q_1 t + q_2 t^2 + q_3 t^3$$

$$p(t) + q(t) = (p_0 + q_0) + (p_1 + q_1)t + (p_2 + q_2)t^2 + (p_3 + q_3)t^3$$

The function above is also a cubic polynomial, so this no escape property also holds.
It can be shown from the properties of scalar multiplication and addition that the remaining properties of vector spaces hold for the set of all $p(t)$:
Commutativity: $p(t) + q(t) = q(t) + p(t)$
Associativity of vector addition: $(p(t) + q(t)) + r(t) = p(t) + (q(t) + r(t))$
Additive identity: There exists 0 in the set of all $p(t)$ such that for all $p(t)$, $0 + p(t) = p(t) + 0 = p(t)$
Existence of inverse: For every $p(t)$, there is $-p(t) \in \mathbb{V}$ such that $p(t) + -p(t) = 0$
Associativity of scalar multiplication: $c(d(p(t))) = (cd)p(t)$
Distributivity of scalar sums: $(c+d)p(t) = cp(t) + dp(t)$
Distributivity of vector sums: $c(p(t) + q(t)) = cp(t) + cq(t)$
Scalar multiplication identity: $1p(t) = p(t)$
Since all properties of a vector space hold, the set of cubic polynomials is a vector space.

(b) Consider the set of real-valued monomials given below:

$$\varphi_0(t) = 1, \quad \varphi_1(t) = t, \quad \varphi_2(t) = t^2, \quad \varphi_3(t) = t^3,$$

where $t \in \mathbb{R}$. Show that every real-valued polynomial of the form

$$p(t) = p_0 + p_1 t + p_2 t^2 + p_3 t^3$$

can be written as a linear combination of the monomials $\varphi_0(t)$, $\varphi_1(t)$, $\varphi_2(t)$, and $\varphi_3(t)$. Would it be accurate to say that these constitute a basis for degree ≤ 3 polynomials? Why or why not? Recall that a basis is a set of linearly independent vectors which span a particular vector space.

Solution: Replacing $1, t, t^2, t^3$ with $\varphi_0(t)$, $\varphi_1(t)$, $\varphi_2(t)$, and $\varphi_3(t)$ in the polynomial above, we get that every polynomial in this form can be written as:

$$p_0 + p_1 t + p_2 t^2 + p_3 t^3 = p_0 \varphi_0(t) + p_1 \varphi_1(t) + p_2 \varphi_2(t) + p_3 \varphi_3(t).$$

So we see that every polynomial in this form can be written as a linear combination of $\varphi_0(t)$, $\varphi_1(t)$, $\varphi_2(t)$, and $\varphi_3(t)$.
Further, since no non-zero linear combination of $\varphi_0(t)$, $\varphi_1(t)$, $\varphi_2(t)$, and $\varphi_3(t)$ exists which gives the zero function ($f(t) = 0 \forall t$), we can know that these are linearly independent. This should be intuitive and there is no need to justify this further; but you can do so by taking the Wronskian (though this is *not* in scope for this class). Since we have a set of linearly independent functions which span the third degree polynomials, it is accurate to call this a basis!

(c) Show that the set of all **derivatives** of polynomials of the form

$$p(t) = p_0 + p_1 t + p_2 t^2 + p_3 t^3$$

forms its own vector space. Call this vector space $\mathbb{D}$.
Note: Simply recognizing that the derivatives lie in the vector space $\mathbb{V}$ from part (a) is insufficient; though if you use (and justify) this fact, it could lead you to a more-concise solution.

Solution:
We recognize that the derivative of our polynomial comes as follows:
$r(t) = \frac{d}{dt} p(t) = p_1 + 2p_2 t + 3p_3 t^2$.
No escape (scaling) property:

$$\alpha r(t) = \alpha p_1 + 2\alpha p_2 t + 3\alpha p_3 t^2 = \alpha(p_1 + 2p_2 t + 3p_3 t^2) = \alpha \frac{d}{dt} p(t)$$

The function above could be the derivative of the function $\alpha p(t)$, so the no escape property holds.

No escape (addition) property:
Let us define two of our derivative polynomials:

$$r(t) = p_1 + 2p_2 t + 3p_3 t^2$$
$$s(t) = q_1 + 2q_2 t + 3q_3 t^2$$

$$r(t) + s(t) = (p_1 + q_1) + 2(p_2 + q_2)t + 3(p_3 + q_3)t^2$$

The function above could be the derivative of a cubic polynomial with coefficients c, $(p_1 + q_1)$, $(p_2 + q_2)$, and $(p_3 + q_3)$ for $1, t, t^2, t^3$ respectively, where c is an arbitrary scalar. So this no escape property also holds.

As with part (a), it can be shown from the properties of scalar multiplication and addition that the remaining properties of vector spaces hold for the set of all derivatives polynomials in $\mathbb{D}$. Since all properties of a vector space hold, the set of derivatives of cubic polynomials is a vector space.

Another possible solution is to note that the set of derivatives of cubic polynomials is a subset of cubic polynomials, i.e. it is a subset of $\mathbb{V}$. Therefore, we can just check subspace conditions rather than all of the vector space conditions, since a subspace of a vector space is a vector space. In this case, we just need to check the no escape properties and that the set has the 0 polynomial in it.

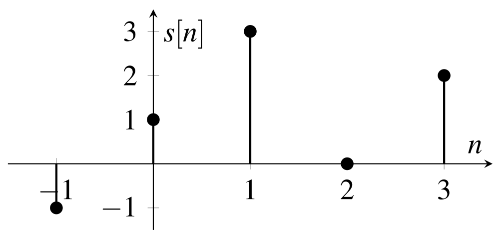
7. Signal Bases

We learned that any discrete-time signal can be written as a sum of scaled and shifted delta functions, namely

$$x[n] = \sum_{m=-\infty}^{\infty} x[m]\delta[n-m], \quad n \in \mathbb{Z}, \tag{1}$$

This is not the only useful way to rewrite a signal. In fact, there are infinitely many such "basis functions" into which a signal may be decomposed. A popular choice is the Fourier basis, which you will learn later when we cover DTFS. For now, we will work with a simpler example.

(a) First, we will practice decomposing a signal using deltas. Write the signal $s[n]$ plotted below in the form of (1). You may assume that the signal is zero outside the pictured region.



Solution:

$$s[n] = -\delta[n+1] + \delta[n] + 3\delta[n-1] + 2\delta[n-3] \tag{2}$$

(b) Recall that the discrete-time unit step function is defined as

$$u[n] = \begin{cases} 0 & \text{if } n < 0, \\ 1 & \text{if } n \geq 0. \end{cases} \tag{3}$$

Write $\delta[n]$ as a linear combination of shifted unit step functions. In other words, use signals of the form $u[n-m]$.

Solution:

$$\delta[n] = u[n] - u[n-1] \tag{4}$$

(c) Now write the signal $s[n]$ from part (a) in terms of shifted unit step functions. Try to do so with as few terms as possible. Then justify why any discrete-time signal can be written as a linear combination of signals of the form $u[n-m]$.

Solution: Plugging (4) into (1), we find

$$x[n] = \sum_{m=-\infty}^{\infty} x[m]\big(u[n-m] - u[n-m-1]\big). \tag{5}$$

Intuitively, since $x[n]$ is a sum of impulses, and an impulse can be expressed as a difference of unit steps, it should make sense that $x[n]$ can also be written in terms of unit steps. Applying this to (2), we get

$$s[n] = -\big(u[n+1] - u[n]\big) + \big(u[n] - u[n-1]\big) + 3\big(u[n-1] - u[n-2]\big) + 2\big(u[n-3] - u[n-4]\big). \tag{6}$$

Equivalently, we can write the signal more compactly as

$$s[n] = -u[n+1] + 2u[n] + 2u[n-1] - 3u[n-2] + 2u[n-3] - 2u[n-4]. \tag{7}$$

(d) Is a decomposition of a signal into shifted unit steps unique? If your answer is "yes," explain why. Otherwise, come up with a restriction on the terms $u[n-m]$ so that the decomposition is unique. Hearken back to your linear algebra class: using this conclusion and the result from part (c), what term best describes this set of unit step signals?

Solution: Even though (6) and (7) look different, they are *mathematically equivalent*! This demonstrates that such a decomposition into unit steps *is* unique. Here is one justification: each unit step provides a change at its value of m and is otherwise constant; for example, $u[n-2]$ only changes at $n=2$). This means that each $u[n-m]$ in a sense tells you how the signal changes at m. In discrete time, there is only one way that the signal can change its value at each point in its domain; for example, it doesn't make sense for a signal to change twice at n=2! Hence, the decomposition is unique.

Part (c) shows that the signals $u[n-m]$ span the space of all discrete-time signals. We just explained that the decomposition into unit steps is unique, so these signals are also linearly independent. This set is therefore a basis for the vector space of all discrete-time signals.

8. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.